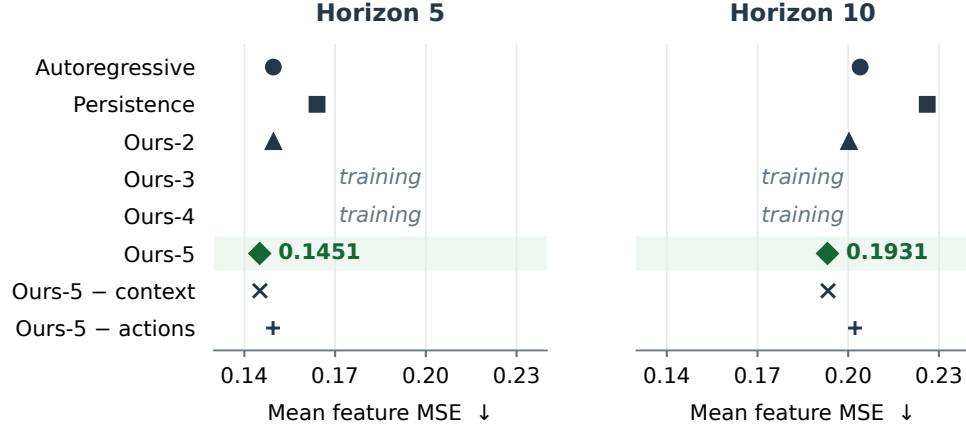


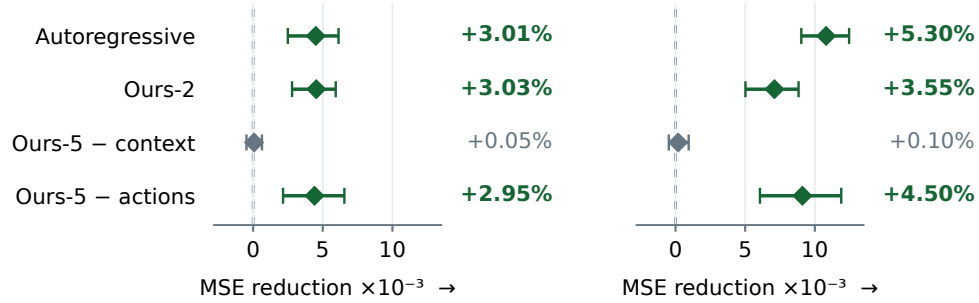
ID	Mechanism	Status
Ours-1	Original context-based ShiftWM	Completed; different protocol
Ours-2	Observed anchor + additive innovation	Completed
Ours-3	Observed anchor + bounded innovation	Training (snapshot)
Ours-4	Gated mixing + unbounded innovation	Training (snapshot)
Ours-5	Gated mixing + bounded innovation	Completed

Table 1: Stable mechanism identifiers, not a performance ranking. Ours-1 uses the original context-based protocol and is excluded from the spatial comparison axis. Context-off and action-free are Ours-5 ablations; autoregression and persistence retain baseline labels. Training statuses refer to this display snapshot.

(a) Native spatial error



(b) Ours-5 gains over trained comparators



Bars: 95% paired CI. Labels: relative point gains.

Figure 1: Stable spatial version comparison on original DROID validation. (a) Absolute native 4×4 standardized feature MSE at query endpoints 5 and 10; dots are equal-episode/equal-seed means. Ours-3/4 have no plotted values while their registered full studies train. (b) Paired MSE reductions of Ours-5 against the four registered trained comparators; right of zero favors Ours-5. Bars are 95% paired recording-session $\times$ training-seed bootstrap intervals (10,000 draws; seed 173); percentages are relative point reductions. Gray intervals include zero; green intervals exclude zero in the favorable direction, without multiplicity adjustment. All 15 completed models trained 30 epochs on identical data; evaluation has 1,631 windows, 141 episodes and 59 sessions with 3 training seeds. Windows are averaged within episodes before equal episode/seed aggregation. Ours-1 uses a different protocol and is not numerically combined here. Versions identify mechanisms, not ranks. The complete companion table retains all 16 native/coarse contrasts, including both unfavorable coarse context-ablation point estimates. These development comparisons establish neither independent-test generalization nor state-of-the-art superiority.

Metric	Comparator	h	Relative gain	MSE reduction $\times 10^3$ [95% CI]
Native	Autoregressive	5	+3.008%	+4.499 [+2.491, +6.132]
Native	Autoregressive	10	+5.300%	+10.809 [+9.028, +12.461]
Native	Ours-2	5	+3.029%	+4.532 [+2.795, +5.930]
Native	Ours-2	10	+3.547%	+7.102 [+5.019, +8.831]
Native	Ours-5 – context	5	+0.048%	+0.069 [-0.478, +0.633]
Native	Ours-5 – context	10	+0.097%	+0.188 [-0.485, +0.943]
Native	Ours-5 – actions	5	+2.948%	+4.406 [+2.149, +6.551]
Native	Ours-5 – actions	10	+4.503%	+9.107 [+6.061, +11.896]
Original 2×2	Autoregressive	5	+0.441%	+0.656 [-1.918, +2.499]
Original 2×2	Autoregressive	10	+2.538%	+5.150 [+2.727, +7.196]
Original 2×2	Ours-2	5	+0.527%	+0.784 [-1.370, +2.281]
Original 2×2	Ours-2	10	+0.472%	+0.938 [-1.784, +2.897]
Original 2×2	Ours-5 – context	5	-0.194%	-0.286 [-1.016, +0.386]
Original 2×2	Ours-5 – context	10	-0.108%	-0.213 [-1.075, +0.644]
Original 2×2	Ours-5 – actions	5	+2.948%	+4.493 [+1.778, +7.086]
Original 2×2	Ours-5 – actions	10	+4.085%	+8.422 [+4.691, +11.892]

Table 2: All 16 completed spatial contrasts: Ours-5 versus each comparator. Positive reductions favor Ours-5; bold green percentages indicate positive point estimates, not statistical significance. Paired 95% recording-session and seed bootstrap intervals use 10,000 draws (seed173), without multiplicity correction. Both unfavorable coarse-coordinate context-ablation point estimates remain included. Native and original 2×2 coordinate errors are different metrics and cannot be pooled. Original validation only.